FIRST PROJECT REPORT

Real Estate Analysis In Paris

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As Part of the Continuing Education Program
Data Engineer Training

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1. Project Overview and Data Exploration

The Real Estate Analysis in Paris project aims to investigate how urban characteristics, particularly green spaces, influence property prices across the city. While the primary focus is on real estate transactions, environmental and urban factors, such as the distribution of parks and risk zones, are recognized as key contributors to price variations.

The goal of Step 1 was to explore unstructured data, define the project context and scope, and take charge of different data sources. The expected deliverable was a comprehensive report summarizing each data source with examples of collected data. This step was crucial to ensure that the subsequent analysis and database design are based on high-quality, compatible datasets.

During Step 1, we identified and analyzed five complementary data sources. The DVF 2022 dataset provided over 45,000 clean real estate transactions in Paris after extensive cleaning, highlighting the need to remove incomplete records and irrelevant columns. The Rent Control dataset offered 10,000 legal reference prices across 80 neighborhoods. Risk Sectors data mapped 1,653 flood-prone zones regulated by the PPRI, while Open Spaces data included 7,268 green areas designated for vegetation. Finally, the opendata.paris API enabled programmatic access to additional real-time datasets.

We found that data quality varies significantly, with the DVF dataset requiring heavy preprocessing, yet all datasets are highly compatible thanks to shared spatial references, allowing future integration. Preliminary exploration also revealed significant geographic variation in rents, ranging from €14.30 to €41.80 per m², and identified updates in the PLU Bioclimatique 2024 that required mapping new dataset endpoints.

The next step (Step 2) focuses on data modeling and database design. We will create a normalized schema (3NF / Snowflake) incorporating as much information as possible from the collected datasets. Deliverables will include a UML diagram, an implemented database, and SQL queries to support subsequent analyses of Paris's real estate market.

2. DVF 2022 — Property Transactions Dataset

DVF & opendata.paris Datasets

This section presents the main real estate transaction data and the complementary datasets fetched via the opendata.paris API.

1.1 Overview

DVF (Demandes de Valeurs Foncières) is the official French government open data on real estate transactions. It contains all property sales in France with geographic coordinates and detailed property information.

Source	**data.gouv.fr**	
URL	https://www.data.gouv.fr/fr/datasets/demandes-de-valeurs-foncieres-geolocalisees/	
Year	**2022**	
Format	**Compressed**	**CSV (~150 MB)**

1.2 Initial Exploration

Metric	Value
Total records (France)	4,676,187
Columns	40
Memory size	~5 GB
Missing values	48.91% of cells
Columns with >50% missing	19

1.3 Data Cleaning Strategy

Following project requirements, three cleaning steps were applied:

Step 1: Filter Paris only (department 75)

Before	After	Reduction
4,676,187 rows	**98,308 rows**	**97.90%**

Step 2: Remove columns with >50% missing values

- ➔ 19 columns removed
- ➔ Examples: ancien_code_commune, lot_surface_carrez, surface_terrain

Step 3: Remove rows with any remaining missing values

Before	After	Reduction
98,308 rows	**45,207 rows**	**54.01%**

1.4 Final Clean Dataset

Metric	Value
Records	**45,207 transactions**
Columns	**21 attributes**
Memory	**31.73 MB (from 5 GB)**
Missing values	**0**
Memory reduction	**99.4%**

1.5 Critical Observation

The column **surface_reelle_bati** (actual built surface area) had **48.83% missing values** — just below the 50% threshold. Although kept, this single column caused ~54% row loss when removing incomplete records.

Recommendation: For Step 2 (Database Modeling), we should consider imputation strategies for this critical attribute.

1.6 Main Columns Retained

Column	Description
id_mutation	**Unique transaction identifier**
date_mutation	**Transaction date**
valeur_fonciere	**Property value (€)**
code_postal	**Postal code**

nom_commune	**City name**
code_departement	**Department code (75 for Paris)**
nombre_lots	**Number of lots**
latitude / longitude	**Geographic coordinates**
surface_reelle_bati	**Built surface area (m²)**
nombre_pieces_principales	**Number of main rooms**

3. opendata.paris API Integration

2.1 Why Use an API?

Instead of manually downloading files, we integrated directly with the opendata.paris REST API. This approach provides:

- ➔ **Automated data retrieval** — no manual downloads
- ➔ **Up-to-date data** — always the latest version
- ➔ **Flexibility** — can filter, sort, and paginate
- ➔ **Professional standard** — industry best practice

2.2 API Configuration

Base URL	**https://opendata.paris.fr/api/explore/v2.1/catalog/datasets/**
Format	**REST API returning JSON**
Authentication	**None required (public API)**
Pagination	**100 records per request**

2.3 Datasets Fetched

We successfully fetched **3 complementary datasets** with a total of **18,921 records**:

Dataset A: Rent Control

Dataset ID	logement-encadrement-des-loyers
Records	10,000 (API limit)
Columns	14
Coverage	80 Paris neighborhoods
Reference rent range	€14.30 - €41.80 per m²
Average rent	€26.40 per m²

Dataset B: Risk Sectors (Flood Risk)

Columns	7	
Source	PLU 2024 (PPRI)	Bioclimatique
	Dataset ID	plub_ppri:
	Records	1,653

Dataset C: Open Spaces (Green Spaces)

Dataset ID	plub_elpv	
Records	7,268 records	green space

Columns	**10**
Source	**PLU Bioclimatique 2024 (ELPV)**

2.4 Technical Challenges & Solutions

Challenge 1: Old Dataset IDs Returned HTTP 404

The previously documented dataset IDs returned HTTP 404 errors.

Reason: The City of Paris adopted the new PLU Bioclimatique on November 20, 2024, which restructured the dataset identifiers.

Solution: Researched the new endpoints:

Status	Old Dataset ID	New Dataset ID
Risk Sectors	**plu-secteurs-de-risques-delimites-par-le-ppri**	**plub_pprizone**
Open Spaces	**plu-espaces-libres-a-vegetaliser-elv**	**plub_elpv**

Challenge 2: Complex Geographic Data Types

Some columns (e.g., geometry) contained nested GeoJSON dictionaries that broke standard pandas operations.

Solution: Implemented defensive programming with try/except blocks to handle complex data types gracefully.

Challenge 3: API Pagination

The API limits results to 100 records per request.

Solution: Built a custom Python function *fetch_opendata_paris()* that automatically handles pagination, progress tracking, and error recovery.

4. Data Quality Assessment Summary

Dataset	Records	Quality	Notes
DVF 2022 (Paris, cleaned)	**45,207**	**Excellent**	**0 missing values**
Rent Control	**10,000**	**Excellent**	**Complete coverage**
Risk Sectors	**1,653**	**Good**	**GeoJSON requires handling**
Open Spaces	**7,268**	**Good**	**GeoJSON requires handling**

TOTAL	**64,128**	**High**	**Ready for Step 2**

5. Technical Stack

Component	Technology
Language	**Python 3.13**
Data manipulation	**pandas, numpy**
API client	**requests**
Visualization	**matplotlib, seaborn**

Notebooks	**Jupyter**
Version Control	**Git + GitHub**
Editor	**Visual Studio Code**

6. Code Repository

All analysis code is documented and version-controlled on GitHub:

Repository: https://github.com/ahmed4maala-afk/real-estate-paris

Key Notebooks

File	Purpose
notebooks/01_data_exploration.ipynb	**DVF dataset analysis & cleaning**

notebooks/02_opendata_paris_api.ipynb	**API integration & 3 datasets**

Project Structure

```
real-estate-paris/
│
├── data/
│   ├── raw/            # Original datasets (gitignored)
│   └── processed/      # Cleaned datasets (gitignored)
│
├── docs/               # Team reports (Tasks A, B, C)
├── notebooks/          # Jupyter analysis notebooks
├── reports/            # Final reports
├── scripts/            # Python scripts
│
```

```
├──                    .gitignore

├──                    README.md

└── requirements.txt
```

7. Insights & Recommendations for Step 2

Key Insights

1. **All datasets share geographic references** - they can be joined on neighborhood code, postal code, or coordinates.

2. **Rent prices vary 3× across Paris** - strong indicator of neighborhood-level segmentation.

3. **Environmental data is regulatory in nature** - PLU Bioclimatique provides legal framework, useful for premium pricing analysis.

Recommendations for Database Modeling (Step 2)

- **Use neighborhood code as primary spatial key** (e.g., code_postal, code_quartier)
- **Extract latitude/longitude as separate columns** for spatial queries
- **Consider PostGIS** (PostgreSQL extension) for geographic data
- **Design a Star Schema** with:
    - **Fact table:** Property transactions (DVF)
    - **Dimension tables:** Neighborhood, Time, Risk Zone, Green Space, Rent Reference

8. ‘Risk Sectors’ Dataset

For the project **“Real Estate Analysis in Paris”**, several datasets about environmental and urban risk zones are used. These datasets help identify areas in Paris that may be affected by flooding, underground instability, or heavy rainfall. Such risks are important because they can directly influence property prices, insurance costs, construction projects, and investment decisions.

The “Risk Sectors” category includes the following datasets:

➔ **PLU bioclimatique – Secteurs de risque – Carrières**
This dataset shows areas with former underground quarries and cavities in Paris. These zones may create ground instability, land subsidence, or structural damage to buildings.

➔ **PLU bioclimatique – Secteurs de risque – PPRI / Flood Risk**
This dataset identifies floodrisk areas along the Seine River and other vulnerable zones. These sectors are considered highrisk areas during flooding events and are regulated by the Flood Risk Prevention Plan (PPRI).

➔ **Zonage pluvial**
This dataset contains information about stormwater and heavy rainfall risk zones in Paris. It shows areas where intense rain may cause flooding or problems in the drainage and sewage systems.

The datasets contain geographic information such as risk zones, surface areas, coordinates, arrondissement numbers, and descriptions of each type of risk. The datasets include multiple mapped geographic records across Paris.

These datasets are very useful for our real estate analysis project because environmental and flood risks can strongly affect the safety, attractiveness, and value of properties. Buyers and investors usually prefer properties located in safer areas with lower environmental risk. In addition, the datasets help compare safe and vulnerable neighborhoods in Paris and support better real estate investment decisions.

7.1 Main Columns in the Dataset

Column Name	Meaning
gid	Unique identifier for each geographic zone
objectid	Internal object number
zone	Type or category of floodrisk zone
libelle	Name or description of the area
surface	Surface area of the zone
arrondissement	Paris district number
geometry	Geographic polygon data used for maps
geo_point_2d	Geographic coordinates (latitude/longitude)
type_risque	Type of environmental risk

9. Rent Control

The dataset contains information about rent control regulations in Paris for the year 2024. It provides official reference rents that are used to regulate housing prices in different neighborhoods of the city. The data is organized according to several important criteria, including geographic sector, neighborhood, apartment size, construction period, and type of rental. Each row in the dataset represents a specific category of housing and the corresponding legal rent limits per square meter.

The column **"Année" (Year)** indicates the year in which the rent regulation applies. **"Secteurs géographiques" (Geographic sectors)** identifies the larger geographic area or sector within Paris, while **"Numéro du quartier" (Neighborhood number)** and **"Nom du quartier" (Neighborhood name)** provide the official code and name of the neighborhood. These columns help users compare rental prices between different parts of the city.

The column **"Nombre de pièces principales" (Number of main rooms)** describes the number of main rooms in the apartment, which is an important factor in determining rent levels. **"Epoque de construction" (Construction period)** specifies the building period or age category of the property, since older and newer buildings may have different rent references. **"Type de location" (Type of rental)** indicates whether the apartment is furnished or unfurnished.

Three key columns define the legal rent values. **"Loyers de référence" (Reference rents)** gives the standard reference rent per square meter. **"Loyers de référence majorés" (Maximum reference rents / Increased reference rents)** shows the maximum legal rent that landlords are generally allowed to charge, while **"Loyers de référence minorés" (Minimum reference rents / Reduced reference rents)** represents the lower reference rent value. These indicators are essential for ensuring compliance with Paris rent control laws.

The dataset also contains geographic information. **"Ville" (City)** identifies the city, which is Paris in this case. **"Numéro INSEE du quartier" (INSEE neighborhood number)** provides the official administrative code assigned to the neighborhood by the French statistical office (INSEE). Finally, **"geo_shape" (Geographic shape)** and **"geo_point_2d" (2D geographic point)** contain geographic coordinates and shapes that can be used for mapping and GIS analysis.

Overall, this dataset is useful for tenants, landlords, researchers, and policymakers who want to analyze housing prices, compare neighborhoods, or study the impact of rent regulation policies in Paris.

10. Open Spaces to be Planted

10.1 Green Spaces Dataset (Paris Open Data)

The Green Spaces Dataset describes all public green areas in Paris such as parks, gardens, and recreational spaces. Each record represents a specific site and includes identification, location, size, management, and historical information. The dataset is useful for urban planning, environmental analysis, and mapping green infrastructure across the city.

Column descriptions

Column	Meaning
Identifiant espace vert	Unique identifier for each green space
Nom de l'espace vert	Name of the green space
Typologie d'espace vert	Type of green space (park, garden, square, etc.)
Catégorie	Functional category of the green space
Adresse Numéro	Street number
Adresse Complément	Additional address information
Adresse type voie	Type of street (street, avenue, etc.)
Adresse Libellé voie	Street name
Code postal	Postal code of the location
Surface calculée	Calculated surface area
Superficie totale réelle	Real total surface area
Surface horticole	Area used for horticulture/planting
Présence cloture	Indicates whether the area is fenced
Périmètre	Perimeter of the green space
Année de l'ouverture	Year the space was opened
Année de rénovation	Year of last renovation
Ancien nom de l'espace vert	Previous name of the site
Année de changement de nom	Year when the name was changed
Nombre d'entités	Number of internal units or sections
Ouverture 24h_24h	Indicates 24hour accessibility
ID_DIVISION	Administrative division ID
ID_ATELIER_HORTICOLE	Horticultural workshop ID
IDA3D_ENB	3D spatial identifier
SITE_VILLES	City site identifier

ID_EQPT	Equipment or facility ID
Compétence	Responsible authority/management unit
Geo Shape	Geographic polygon (map geometry)
URL_PLAN	Link to map or plan of the site
Geo point	Geographic coordinates (center point)
last_edited_user	User who last modified the record
last_edited_date	Date of last modification

Short explanation

This dataset provides a detailed inventory of green spaces in Paris, including their names, locations, physical characteristics, and administrative details. It allows users to analyze the distribution and size of parks and gardens across different parts of the city. Address related fields help precisely locate each green space, while surface and perimeter data describe their physical dimensions. Historical fields such as opening year and renovation year allow tracking of changes over time. Administrative identifiers show how each site is managed within the city system. Geographic fields like geometry and coordinates enable spatial mapping and GIS analysis. Overall, the dataset supports urban planning, environmental monitoring, and public space management in Paris.